Waypoint-to-State/Control Pipelines:
2,000-Sample Single-Score Findings

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Purpose and common calibration contract

Five waypoint-to-state/control pipelines were evaluated on the same 2,000 actual refined and verified RRT/LLM pairs, sample IDs 0–1999 from `conformal_rrt_calibration_dataset_2001.csv`. Every generated row was used. The plant and controller were held fixed:

$$\dot{x} = Ax + Bu, \qquad u = -K(x - \hat{x}_d) + \hat{u}_d,$$

where $x = [p_x, p_y, v_x, v_y]^\top$ and the damped velocity dynamics are

$$\dot{p} = v, \qquad \dot{v} = -1.1v + 1.1u.$$

For expert references (x_d, u_d) and LLM references $(\hat{x}_d, \hat{u}_d)$, the disturbance entering the expert-relative closed-loop error dynamics is

$$w(t) = B\big(\hat{u}_d(t) - u_d(t)\big) + BK\big(\hat{x}_d(t) - x_d(t)\big).$$

The single trajectory-level nonconformity score is

$$S_i = \max_t \sqrt{w_i(t)^\top P w_i(t)}.$$

At coverage $1 - \delta = 0.90$ and $N = 2000$, the conformal order-statistic rank is

$$k = \lceil (1 - \delta)(N + 1) \rceil = \lceil 0.9(2001) \rceil = 1801.$$

Measured results

The CARE metric gives

$$\alpha_P = 1.096230, \qquad \lambda_{\min}(P) = 0.612622,$$

and the common asymptotic Euclidean state-radius calculation is

$$\rho_\infty = \frac{q_w}{\alpha_P \sqrt{\lambda_{\min}(P)}}.$$

Pipeline	q_w	ρ_∞ [m]
Minimum-control QP, shared clock	**1.641463**	**1.913080**
Guided linear/parabolic fit, shared clock	2.364193	2.755402
Minimum snap, shared clock	2.689805	3.134893
Minimum snap, independent clocks	2.982504	3.476026
Minimum snap, Frenet source trajectories	2.982504	3.476026

All five CSVs contained exactly 2,000 rows, the same ordered sample IDs and verified waypoint pairs, finite four-state/two-input samples, and a stored quantile equal to the independently recomputed rank-1801 value. Exactly 1,801 rows were at or below each reported quantile. The existing offline conformal verifier completed for every file; on representative row 998 it accepted the score and placed the simulated closed-loop trajectory inside the corresponding tube.

Why the shared-clock QP is retained

The QP enforces the verified waypoints as equality constraints under the repository's exact discrete damped double-integrator dynamics and minimizes the squared control norm. Unlike a seventh-order polynomial fit, it does not obtain control by differentiating high-order position coefficients. Shared total duration also prevents the RRT and LLM paths from being compared at incompatible terminal clocks. In this population those two choices produced the smallest combined disturbance quantile and therefore the smallest radius under the same P, K, α_P, coverage, and sample count.

The Frenet dataset changes the legacy progress-aligned s_x, s_u diagnostics, but its stored minimum-snap trajectories are the same trajectories as the independent-clock dataset. The certified single score above is defined in physical time because it enters $\dot{e} = (A - BK)e + w$. Consequently the two time-domain S_i populations, quantiles, and radii are identical. Substituting a progress-indexed score into this time-domain bound without re-deriving the chain-rule dynamics would not be a valid certificate.

Retained numerical certificate

For the winning pipeline,

$$q_w = 1.641463, \qquad \rho_\infty = \frac{1.641463}{1.096230\sqrt{0.612622}} = 1.913080 \text{ m}.$$

With initial expert-relative error $e(0)$, the complete comparison bound is

$$\rho(t) = \sqrt{\frac{\lambda_{\max}(P)}{\lambda_{\min}(P)}}\|e(0)\|_2 e^{-\alpha_P t} + \rho_\infty\big(1 - e^{-\alpha_P t}\big).$$

This certificate concerns the calibrated planar reference discrepancy and damped model. It is not, by itself, a certificate for the full nonlinear quadrotor, actuator saturation, estimator failures, communications loss, or unmodelled PX4 inner-loop behavior.

Conclusion

Under one score, one 90% failure budget, and one certified metric, the shared-clock minimum-control QP is the least-conservative tested waypoint-to-state/control pipeline. It is therefore the sole trajectory generator retained by the cleaned runtime and calibration workflow; the complete multi-method implementation and data remain archived on the `all_pipelines_tested` branch.